Statistics for Data Science - 2

Week 8 Graded assignment

1. Let $X_1, X_2, \ldots, X_{50} \sim$ i.i.d. Poisson(0.04) and let $Y = \sum_{i=1}^{50} X_i$. Use Central Limit theorem to find $P(Y > 3)$. Enter the answer correct to 2 decimal places.

   Solution:

   Let $X \sim \text{Poisson}(0.04)$.
   Consider the samples $X_1, X_2, \ldots, X_{50}$ from X.
   $E[X] = \text{Var}[X] = 0.04$
   $E[Y] = E\left[\sum_{i=1}^{50} X_i\right] = 50 \times 0.04 = 2$, $\text{Var}[Y] = \text{Var}\left[\sum_{i=1}^{50} X_i\right] = 50 \times 0.04 = 2$
   To find: $P(Y > 3)$.
   By CLT, we know that

$$\frac{Y - n\mu}{\sigma\sqrt{n}} \sim \text{Normal}(0,1)$$
$$\Rightarrow \left(\frac{Y-2}{\sqrt{2}}\right) \sim \text{Normal}(0,1)$$

   Now,

$$\begin{aligned} P(Y > 3) &= P(Y - 2 > 1) \\ &= P\left(\frac{Y-2}{\sqrt{2}} > \frac{3-2}{\sqrt{2}}\right) \\ &= P(Z > 0.707) \\ &= 1 - F_Z(0.707) = 1 - 0.76 = 0.24 \end{aligned}$$

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2. Let the moment generating function of a random variable X be given by

$$M_X(\lambda) = \left(\frac{1}{4}\right)e^{-2\lambda} + \left(\frac{1}{40}\right) + \left(\frac{3}{10}\right)e^{-\lambda} + \left(\frac{3}{40}\right)e^{2\lambda} + \left(\frac{7}{20}\right)e^{\lambda}$$

   Find the distribution of X.

X	-2	-1	0	1	2
$P(X = x)$	$\frac{1}{4}$	$\frac{3}{40}$	$\frac{3}{10}$	$\frac{1}{40}$	$\frac{7}{20}$

   (a)

X	-2	-1	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{40}$	$\frac{3}{10}$	$\frac{3}{40}$	$\frac{7}{20}$

(b)

X	-2	-1	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{3}{10}$	$\frac{1}{40}$	$\frac{7}{20}$	$\frac{3}{40}$

(c)

X	-2	-1	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{3}{40}$	$\frac{1}{40}$	$\frac{7}{20}$	$\frac{3}{10}$

(d)

Solution:

The MGF of a discrete random variable X with the PMF $f_X(x) = P(X = x)$, $x \in T_X$ is given by

$$\begin{aligned} M_X(\lambda) &= E[e^{\lambda X}] \\ &= \sum_{x \in T_X} P(X = x) e^{\lambda x} \end{aligned}$$

Now, MGF of a random variable X is given as

$$M_X(\lambda) = \left(\frac{1}{4}\right) e^{-2\lambda} + \left(\frac{1}{40}\right) + \left(\frac{3}{10}\right) e^{-\lambda} + \left(\frac{3}{40}\right) e^{2\lambda} + \left(\frac{7}{20}\right) e^{\lambda}$$

Therefore, distribution of X is given by

X	-2	-1	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{3}{10}$	$\frac{1}{40}$	$\frac{7}{20}$	$\frac{3}{40}$

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3. A fair coin is tossed 1000 times. Use CLT to compute the probability that head appears at most 520 times. Enter the answer correct to 3 decimal places.

Solution:

Define a random variable X such that

$$X = \begin{cases} 1 & \text{if head appears on tossing a fair coin} \\ 0 & \text{otherwise} \end{cases}$$

Therefore, $E[X] = \mu = \frac{1}{2}$ and
$\text{Var}(X) = \sigma^2 = \frac{1}{2}.\frac{1}{2} = \frac{1}{4}$

Let $X_1, X_2, \ldots, X_{1000}$ be outcomes on tossing the fair coin 1000 times.
Notice that $X_1 + X_2 + \ldots + X_{1000}$ will denote the number of times head appears in 1000 tosses.

Let $S = X_1 + X_2 + \ldots + X_{1000}$

To find: $P(S \leq 520)$

By CLT, we know that

$$\frac{S - 1000\mu}{\sigma\sqrt{n}} \sim \text{Normal}(0,1)$$
$$\Rightarrow \frac{S - 500}{5\sqrt{10}} \sim \text{Normal}(0,1)$$

Now,

$$\begin{aligned} P(S \leq 520) &= P(S - 500 \leq 20) \\ &= P\left(\frac{S-500}{5\sqrt{10}} \leq \frac{20}{5\sqrt{10}}\right) \\ &= P(Z \leq 1.26) \\ &= 0.896 \end{aligned}$$

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4. Let $X_1, X_2, \ldots, X_{500} \sim$ i.i.d Normal$(0,1)$. Evaluate $P(X_1^2 + X_2^2 + \ldots + X_{500}^2 > 550)$ using Central Limit theorem. Enter the answer correct to 2 decimal places.
Hint: $(X_1^2 + X_2^2 + \ldots + X_{500}^2) \sim$ Gamma $(250, 0.5)$.

Solution:

Given $X_1, \ldots, X_{500} \sim$ i.i.d. Normal$(0,1)$.

We know that if $X \sim \text{Normal}(0,1) \implies X^2 \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right)$

Also, Sum of n independent Gamma(α, β) is Gamma$(n\alpha, \beta)$.

Therefore, $X_i^2 \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right)$, for all i.

and $(X_1^2 + X_2^2 + \ldots + X_{500}^2) \sim$ Gamma $(250, 0.5)$

Let $Y = Y_1 + Y_2 + \ldots + Y_{500}$, where $Y_i = X_i^2$ for all $i : 1 \to 500$

$E[Y_i] = \frac{0.5}{0.5} = 1$ and $\text{Var}[Y_i] = \frac{0.5}{0.25} = 2$, for $i : 1 \to 500$

$E[Y] = \dfrac{250}{0.5} = 500$ and $\text{Var}[Y] = \dfrac{250}{0.5^2} = 1000$

To find: $P(Y > 550)$

By CLT, we know that

$$\frac{Y - 500\mu}{\sigma\sqrt{n}} \sim \text{Normal}(0,1)$$
$$\Rightarrow \frac{Y - 500}{10\sqrt{10}} \sim \text{Normal}(0,1)$$

Now,

$$\begin{aligned} P(Y > 550) &= P(Y - 500 > 50) \\ &= P\left(\frac{Y - 550}{10\sqrt{10}} > \frac{5}{\sqrt{10}}\right) \\ &= P(Z > 1.58) \\ &= 1 - F_Z(1.58) = 1 - 0.94 = 0.06 \end{aligned}$$

Use the below information to answer questions 5 and 6.

Let X be a random variable having the gamma distribution with the parameters $\alpha = 2n$ and $\beta = 1$.
Hint:

- If $X \sim \text{Gamma}(\alpha, \beta), E[X] = \dfrac{\alpha}{\beta}$ and $\text{Var}[X] = \dfrac{\alpha}{\beta^2}$
- Sum of n independent $\text{Gamma}(\alpha, \beta)$ is $\text{Gamma}(n\alpha, \beta)$

5. Use the Weak Law of Large number to find the value of n such that

$$P\left(\left|\frac{X}{2n} - 1\right| > 0.01\right) < 0.01$$

(a) 505000
(b) 470000
(c) 498000
(d) 482000

Solution:

Given $X \sim \text{Gamma}(2n, 1)$
Let $X = X_1 + X_2 + X_3 + \ldots + X_{2n}$, where $X_i \sim \text{Gamma}(1, 1)$.

$E[X] = \mu = 1$ and $\text{Var}(X) = \sigma^2 = 1$
$E[\bar{X}] = 1$ and $\text{Var}[\bar{X}] = \dfrac{1}{2n}$

To find: The value of n such that $P\left(\left|\dfrac{X}{2n} - 1\right| > 0.01\right) < 0.01$.

By weak law of large numbers, we have

$$P(|\overline{X} - \mu| > \delta) \leq \frac{\sigma^2}{n\delta^2}$$
$$\Rightarrow P\left(\left|\frac{X}{2n} - 1\right| > 0.01\right) \leq \frac{1}{2n \times 0.01^2}$$

Therefore, $\dfrac{1}{2n \times 0.01^2} < 0.01 \implies 2n > \dfrac{1}{0.01^3} \implies n > 500000.$

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6. Use CLT to find the value of n such that

$$P\left(\left|\frac{X}{2n} - 1\right| > 0.01\right) < 0.01$$

Hint: Use $F_Z(2.58) = 0.995, F_Z(1.96) = 0.975$ if needed.

(a) 34570
(b) 33500
(c) 32500
(d) 30000

Solution:
$E[X_1 + \ldots + X_{2n}] = 2n$ and $\text{Var}[X_1 + \ldots + X_{2n}] = 2n$

To find: The value of n such that $P\left(\left|\dfrac{X}{2n} - 1\right| > 0.01\right) < 0.01$.

By CLT, we know that

$$\frac{X - 2n\mu}{\sigma\sqrt{n}} \sim \text{Normal}(0, 1)$$
$$\Rightarrow \frac{X - 2n}{\sqrt{2n}} \sim \text{Normal}(0, 1)$$

Now,

$$
\begin{aligned}
P\left(\left|\frac{X}{2n}-1\right|>0.01\right) &< 0.01 \\
\implies P\left(\left|\frac{X_1+\ldots+X_n}{2n}-1\right|>0.01\right) &< 0.01 \\
\implies P\left(\left|\frac{X_1+\ldots+X_n-2n}{\sqrt{2n}}\right|>0.01\sqrt{2n}\right) &< 0.01 \\
\implies P(\mid Z \mid > 0.01\sqrt{2n}) &< 0.01 \\
\implies 2P(Z > 0.01\sqrt{2n}) &< 0.01 \\
\implies 1-F_Z(0.01\sqrt{2n}) &< \frac{0.01}{2} \\
\implies F_Z(0.01\sqrt{2n}) &> 0.995 \\
\implies F_Z(0.01\sqrt{2n}) &> F_Z(2.58) \\
\implies n &> 33282
\end{aligned}
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7. Let the time taken (in hours) for failure of an electric bulb follow the exponential distribution with the parameter 0.05. Suppose that 100 such light bulbs say $L_1, L_2, \ldots, L_{100}$ are used in the following manner: For every i, as soon as the light L_i fails, L_{i+1} becomes operative, where $i : 1 \to 99$ (i.e. If L_1 fails, L_2 becomes operative, if L_2 fails, L_3 becomes operative, and so on). Let the total time of operation of 100 bulbs be denoted by T. Using CLT, compute the probability that T exceeds 2500 hours.

   (a) $F_Z(1.5)$
   (b) $1 - F_Z(1.5)$
   (c) $F_Z(2.5)$
   (d) $1 - F_Z(2.5)$

Solution:

Given, time to failure (in hours) of an electric bulb has the exponential distribution with the parameter $\lambda = 0.05$.
Since, the bulbs are used in such a way, that as soon as light L_1 fails, L_2 becomes operative, L_2 fails, L_3 becomes operative, and so on.

We know that if $X \sim \text{Gamma}(\alpha, \beta)$ with parameter $\alpha = 1$, then $X \sim \text{Exp}(\beta)$.
Also, sum of n i.i.d. $\text{Exp}(\lambda)$ is $\text{Gamma}(n, \lambda)$.

Since each of the L_i's are exponentially distributed with parameter $= 0.05$, therefore

$$L_1 + \ldots + L_{100} \sim \text{Gamma}(n\alpha, \beta) = \text{Gamma}(100, 0.05)$$

Let $T = L_1 + \ldots + L_{100}$

$E[L_i] = \mu = \dfrac{1}{0.05} = 20$ and $\text{SD}[L_i] = \sigma = \dfrac{1}{0.05} = 20$

To find: $P(T \geq 2500)$

By CLT, we know that

$$\frac{T - 100\mu}{\sigma\sqrt{n}} \sim \text{Normal}(0, 1)$$
$$\Rightarrow \frac{T - 2000}{20\sqrt{100}} \sim \text{Normal}(0, 1)$$

Now,

$$\begin{aligned} P(T \geq 2500) &= P(T - 2000 \geq 500) \\ &= P\left(\frac{T - 2000}{200} \geq \frac{500}{200}\right) \\ &= P(Z \geq 2.5) \\ &= 1 - F_Z(2.5) \end{aligned}$$

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8. Suppose speeds of vehicles on a particular road are normally distributed with mean 36 mph and standard deviation 2 mph. Find the probability that the mean speed $\overline{X}$ of 20 randomly selected vehicles is between 35 and 38 mph.

   (a) $F_Z(\sqrt{5}) - F_Z(-\sqrt{5})$
   (b) $F_Z(\sqrt{20}) - F_Z(-\sqrt{20})$
   (c) $F_Z(\sqrt{38}) - F_Z(-\sqrt{35})$
   (d) $F_Z(\sqrt{20}) - F_Z(-\sqrt{5})$

   Solution:

   Let X denote the speed of a vehicle on a particular road.
   Given that $X \sim \text{Normal}(36, 2^2)$.
   Therefore, $\mu = 36$ and $\sigma = 2$
   Select $X_1, X_2, \ldots X_{20}$ samples such that $X_1, X_2, \ldots X_{20} \sim$ iid X

   Let $\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_{20}}{20}$ and $S = X_1 + X_2 + \ldots + X_{20}$

   To find: $P(35 < \overline{X} < 38)$ From CLT, we know that

$$\frac{X_1 + X_2 \ldots + X_n - nE[X]}{\sqrt{n}\sigma} \sim \text{Normal}(0,1)$$
$$\Rightarrow \frac{S - n\mu}{\sqrt{n}\sigma} \sim \text{Normal}(0,1)$$
$$\Rightarrow \frac{(S - 36(20))}{(2\sqrt{20})} \sim \text{Normal}(0,1)$$

Now,

$$\begin{aligned}
P(35 < \overline{X} < 38) &= P(35 < \frac{S}{20} < 38) \\
&= P(-1 < \frac{S}{20} - 36 < 2) \\
&= P(-1 < \frac{S - 36(20)}{20} < 2) \\
&= P(\frac{-\sqrt{20}}{2} < \frac{S - 36(20)}{2\sqrt{20}} < \sqrt{20}) \\
&= P(-\sqrt{5} < \frac{S - 36(20)}{2\sqrt{20}} < \sqrt{20}) \\
&= F_Z(\sqrt{20}) - F_Z(-\sqrt{5})
\end{aligned}$$

9. A company delivers packages, and the delivery time (T) for a package follows an exponential distribution with a mean of 30 minutes. Suppose the company delivers n packages in a day. What is the minimum number of packages n required to ensure that the probability that the average delivery time ($\overline{T}$) for the day lies between 28 and 32 minutes is at least 0.95? Use CLT to approximate.

Solution:

Given, T $\sim$ Exponential(λ), and its mean(μ) is 30 mintutes. So

$$\lambda = \frac{1}{30},$$
$$\mu = 30,$$
$$\sigma = 30$$

We have to find out the minimum number of packages n required to ensure that the probability that the average delivery time ($\overline{T}$) for the day lies between 28 and 32 minutes is at least 0.95. i.e.

$$P(28 < \bar{T} < 30) \geq 0.95 \tag{1}$$

By CLT, we know that

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \text{Normal}(0, 1)$$

We can bring equation 1 in the same form with some manipulation to calculate the value of n, so we get,

$$P\left(\frac{28-30}{30/\sqrt{n}} < \frac{\bar{T}-30}{30/\sqrt{n}} < \frac{28-30}{30/\sqrt{n}}\right) \geq 0.95$$

$$P\left(\frac{28-30}{30/\sqrt{n}} < Z < \frac{28-30}{30/\sqrt{n}}\right) \geq 0.95$$

Where, $Z \sim \text{Normal}(0, 1)$

$$P\left(\frac{-\sqrt{n}}{15} < Z < \frac{\sqrt{n}}{15}\right) \geq 0.95$$

$$1 - 2F_Z\left(\frac{-\sqrt{n}}{15}\right) \geq 0.95$$

$$F_Z\left(\frac{-\sqrt{n}}{15}\right) \leq 0.025$$

$$F_Z\left(\frac{-\sqrt{n}}{15}\right) \leq F_Z(-1.96)$$

$$\frac{-\sqrt{n}}{15} \leq -1.96$$

$$n \geq (1.96 \times 15)^2$$

$$n \geq 865$$

10. Two fair dice are rolled simultaneously. Let Y represent the sum of the numbers appearing on the upper faces of the dice. Determine the moment-generating function (MGF) of $Y - E[Y]$. Select all correct options from below.

    (a) $e^{-3.5t} \cdot \frac{1}{6}\left(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)$

    (b) $\frac{1}{6}\left(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)$

    (c) $[e^{-3.5t} \cdot \frac{1}{6}\left(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)]^2$

    (d) $e^{-7t} \cdot \frac{1}{36}\left(e^{-5t} + 2e^{-4t} + 3e^{-3t} + 4e^{-2t} + 5e^{-t} + 6 + 5e^t + 4e^{2t} + 3e^{3t} + 2e^{4t} + e^{5t}\right)$

Solution:

Here,
X_i = Number that appears on throw of a die, and
$Y = (X_1 + X_2)$
So, $E[X] = 3.5$ and $E[Y] = 7$

We can calculate the MGF of Y in two ways.

(a) We can square the MGF of X to get the MGF of Y (Since, MGF of sum of n iid random variables, X is the nth power of the MGF of X). PMF table for centralized version of X is,

X	-2.5	-1.5	-0.5	0.5	1.5	2.5
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

From here we can get the MGF of X as,

$$M_X(t) = E[e^{\lambda X}] = \frac{1}{6}\left(e^{-2.5t} + e^{-1.5t} + e^{-0.5t} + e^{0.5t} + e^{1.5t} + e^{2.5t}\right)$$

So, the MGF of Y will be,

$$M_Y(t) = M_X(t)^2 = \frac{1}{36}\left(e^{-2.5t} + e^{-1.5t} + e^{-0.5t} + e^{0.5t} + e^{1.5t} + e^{2.5t}\right)^2$$

Which on rearrangement, can be written as,

$$M_Y(t) = M_X(t)^2 = \frac{1}{36} \cdot e^{-7t} \cdot \left(e^{t} + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)^2$$

(b) We can find the PMF table of centralized version of Y and from there we can directly calculate the MGF of Y.
Here's the PMF table of centralized version of Y,

Y	-5	-4	-3	-2	-1	0	1	2	3	4	5
$P(Y = y)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

So, we can write the MGF of Y as

$$M_Y(t) = E[e^{\lambda y}] = \frac{1}{36}e^{-5t} + \frac{2}{36}e^{-4t} + \frac{3}{36}e^{-3t} + \frac{4}{36}e^{-2t} + \frac{5}{36}e^{-t} + \frac{6}{36} + \frac{5}{36}e^{t} + \frac{4}{36}e^{2t} + \frac{3}{36}e^{3t} + \frac{2}{36}e^{4t} + \frac{1}{36}e^{5t}$$

Which on rearrangement, can be written as,

$$M_Y(t) = E[e^{\lambda y}] = e^{-7t} \cdot \frac{1}{36}(e^{2t} + 2e^{3t} + 3e^{4t} + 4e^{5t} + 5e^{6t} + 6e^{7t} + 5e^{8t} + 4e^{9t} + 3e^{10t} + 2e^{11t} + e^{12t})$$

Therefore, option (c) and (d) are right.